End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B Tech CSAI/ECEAI /AI-ML>, Semester:<II>
(July, 2023)

Subject Code: BAS 108	Subject: Probability and Statistics
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)
	(a) Let X be a random variable with MGF $M_X(t)$, and let a, b be constants. If random variable $Y=aX+b$, then find the MGF of Y.	
	(b) For any events A and B with $P(B)>0$, show that $P(A\|B)+P(A'\|B)=1$.	
	(c) If X and Y are independent random variable, then find covariance between $X+Y$ and $X-Y$.	
	(d) If X and Y are discrete random variables. If $var(X)=var(Y)=\sigma^2$, $Cov(X,Y)=\frac{\sigma^2}{2}$, then find $var(2X-3Y)$.	
	(e) Compute the Cumulative distribution function of the Geometric distribution.	
	(f) If $X \sim Bin(11,p)$ such that $P(X=8)=P(X=7)$,then find $var(X)$.	
	(g) Explain null hypothesis and alternative hypothesis.	
	(h) Define the term critical region in terms of hypothesis.	
	UNIT-I	
Q2	Let X be a discrete random variable with the following PMF $f_X(x)=\begin{cases}\frac{1}{4} & ,if\ x=-2\\ \frac{1}{8} & ,if\ x=-1\\ \frac{1}{8} & ,if\ x=0\\ \frac{1}{4} & ,if\ x=1\\ \frac{1}{4} & ,if\ x=2\\ 0, & otherwise\end{cases}$ If random variable $Y=(X+1)^2$, then (i) Find the range of Y. (ii) Find the probability mass function of Y. (iii) Find the expected value and variance of Y.	(2+3+5)
Q3	(a) Four boxes A,B,C,D contain fuses. The boxes contain 5000, 3000, 2000 and 1000 fuses respectively. The percentages of fuses in boxes which are defective are 3%, 2%, 1% and 5% respectively.one fuse in selected at random arbitrarily from one of the boxes. It is found to be defective fuse. Find the probability that it has come from box D. (b) The mean of 16 items was found to be 30. On rechecking, it was found that two items were wrongly taken as 22 and 18 instead of 32 and 28 respectively. Find the correct mean.	(5+5)
	UNIT-II	
Q4	(a) Harry and Jane have agreed to meet for lunch between 1:00 p.m. and 2:00 p.m. Denote Jane's arrival time by X, Harry's by Y, and suppose X and Y are independent with probability density functions: $f_X(x)=\begin{cases}3x^2 & ,if\ 0<x<1\\ 0, & otherwise\end{cases}$ and $f_Y(y)=\begin{cases}2y & ,if\ 0<y<1\\ 0, & otherwise\end{cases}$ Find the probability that Jane arrives before Harry. (b) A random sample of 4 rings is selected from a bag containing 3 silver rings, 2 red rings, and 3 blue rings. Given that X is the number of silver rings and Y is the number of red rings, find the joint probability mass function of X and Y.	(5+5)
Q5	Let X and Y be random variables that have the joint pdf	(2+5+3)

P.T.

$$f_{XY}(x,y) = \begin{cases} c(x+y), & 0 < x < 1;\ 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Find c.
(ii) Find $P(X > Y | X < 1/2)$.
(iii) Find $f_{Y|X}(y|x)$

UNIT-III

Q6 (a) Let $Y = X + 30$ and $Z = 3X - 4$. $X \sim unif(-1,2)$. Find $Cov(Y, Z)$. **(5+5)**

(b) If electricity power failures occur according to a Poisson distribution with an average of 33 failures every twenty weeks, calculate the probability that there will not be more than one failure during a particular week.

Q7 (a) The following information is given: **(5+5)**

	X (in Rs.)	Y (in Rs.)
Arithmetic Mean	40	6
Standard Deviation	10	1.5

Coefficient of correlation between X and Y is 8/15.

(i) Find the regression coefficient of Y on X.
(ii) Find the most likely value of Y when X is Rs. 100.

(b) On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed. What is the probability that a computer part lasts between nine and 11 years?

UNIT-IV

Q8 (a) A sample of 20 items has mean 42 units and standard deviation 5 units. Test the hypothesis that it is a random sample from normal population with mean 45 units. **(5+5)**

(b) In 120 throws of a single die, the following observation of faces was observed.

Face	1	2	3	4	5	6
Frequency	30	25	18	10	22	15

Can you say that the die is biased?

Q9 (a) In one sample of 10 observations the sum of squares of deviation of sample values from the sample mean was 120 and in another sample of 12 observations it was 314. Test whether this difference is significant at 5% level of significance. **(5+5)**

(b) Two random samples are drawn from the normal populations are given below:

Sample 1	17	27	18	25	27	29	13	17
Sample 2	16	16	20	27	26	25	21	

Can we conclude that the two population are drawn from the same population? Test at 5% level of significance.

Cumulative Standard Normal Distribution: *F(-0.87) = 0.19215, F(-1.34) = 0.09012, F(1.34) = 0.90988.*

Critical values of t-distribution: $t_{0.05} = 2.09$(at 19 df, two − tailed), $t_{0.05} = 1.729$(at 19 df, one − tailed), $t_{0.05} = 2.16$(at 13 df, two − tailed), $t_{0.05} = 1.771$(at 13 df, one − tailed)

Critical values of F-distribution (5% significance level): $F_{(9,11)} = 2.9$, $F_{(11,9)} = 3.1$, $F_{(8,7)} = 3.73$

Critical values of χ^2-distribution (5% significance level): $\chi^2_{0.05} = 11.09$ (at 5 df), $\chi^2_{0.05} = 16.919$ (at 9 df).